

BASIC SCIENCES

Interference With ESAM (Endothelial Cell-Selective Adhesion Molecule) Plus Vascular Endothelial-Cadherin Causes Immediate Lethality and Lung-Specific Blood Coagulation

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OBJECTIVE: Vascular endothelial (VE)-cadherin is of dominant importance for the formation and stability of endothelial junctions, yet induced gene inactivation enhances vascular permeability in the lung but does not cause junction rupture. This study aims at identifying the junctional adhesion molecule, which is responsible for preventing endothelial junction rupture in the pulmonary vasculature in the absence of VE-cadherin.

APPROACH AND RESULTS: We have compared the relevance of ESAM (endothelial cell-selective adhesion molecule), JAM (junctional adhesion molecule)-A, PECAM (platelet endothelial cell adhesion molecule)-1, and VE-cadherin for vascular barrier integrity in various mouse tissues. Gene inactivation of ESAM enhanced vascular permeability in the lung but not in the heart, skin, and brain. In contrast, deletion of JAM-A or PECAM-1 did not affect barrier integrity in any of these organs. Blocking VE-cadherin with antibodies caused lethality in ESAM^{-/-} mice within 30 minutes but had no such effect in JAM-A^{-/-}, PECAM-1^{-/-} or wild-type mice. Likewise, induced gene inactivation of VE-cadherin caused rapid lethality only in the absence of ESAM. Ultrastructural analysis revealed that only combined interference with VE-cadherin and ESAM disrupted endothelial junctions and caused massive blood coagulation in the lung. Mechanistically, we could exclude a role of platelet ESAM in coagulation, changes in the expression of other junctional proteins or a contribution of cytoplasmic signaling domains of ESAM.

CONCLUSIONS: Despite well-documented roles of JAM-A and PECAM-1 for the regulation of endothelial junctions, only for ESAM, we detected an essential role for endothelial barrier integrity in a tissue-specific way. In addition, we found that it is ESAM which prevents endothelial junction rupture in the lung when VE-cadherin is absent.

VISUAL OVERVIEW: An online [visual overview](#) is available for this article.

Key Words: blood coagulation ■ cell adhesion molecules ■ endothelial cell ■ lung ■ permeability

The stability and permeability of the blood vessel wall rely largely on endothelial junctions, which are formed and regulated by a multitude of cell adhesion molecules.¹ Vascular endothelial (VE)-cadherin is arguably one of the best studied endothelial-specific adhesion molecules, which is of dominant importance for junction stability.² Like other classical cadherins, it is a highly efficient homophilic adhesion molecule. It is specific for endothelial cells (ECs) and essential for the development of the vasculature,^{3,4}

and its function is regulated by tyrosine phosphorylation during leukocyte extravasation and the induction of vascular permeability in inflammation.^{5,6} Although blocking of VE-cadherin with antibodies destabilizes and disintegrates cell junctions of cultured ECs,⁷ in vivo these antibodies only enhance vascular permeability in the lung and the heart but not in the skin, brain, trachea, or intestine.⁸ Conditional gene inactivation of VE-cadherin had similar results with enhanced permeability in lung and heart but not in skin and

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Nonstandard Abbreviations and Acronyms

CHO	Chinese hamster ovary
ECs	endothelial cells
ESAM	endothelial cell-selective adhesion molecule
FACS	fluorescence-activated cell sorter
JAM-A	junctional adhesion molecule
MLMVECs	mouse lung microvascular endothelial cells
PDZDBM	PDZ-domains binding motifs
PECAM	platelet endothelial cell adhesion molecule
SH3DBM	SH3-domains binding motifs
TNF	tumor necrosis factor
VE-cadherin	vascular endothelial-cadherin
WT	wild type

brain.⁹ Remarkably, mice even survived several weeks without VE-cadherin, despite clear signs of reduced lung and heart function.⁹ Surprisingly, despite strongly enhanced vascular permeability for plasma macromolecules, electron microscopy revealed no sign of ultrastructural rupture of junctions in lung and heart vessels.⁹

Obviously, once endothelial junctions have been properly formed, they can survive the removal of VE-cadherin, and other adhesion molecules are able to maintain their overall structure, and in many organs even their barrier function for large molecules. There are several homophilic adhesion molecules that could be responsible for this, among them, PECAM (platelet endothelial adhesion molecule)-1,¹⁰ the closely related JAM (junctional adhesion molecules)-A, B, and C,¹¹ and ESAM (endothelial cell-selective adhesion molecule) which is related to the JAMs.¹²

PECAM-1 is a member of the Ig superfamily, which is expressed on ECs, platelets and most nonerythroid cells of the hematopoietic lineage.¹³ PECAM-1 is of prominent importance for the transendothelial migration of leukocytes.^{14,15} First indications for the adhesive activity were based on its accumulation at cell contacts of transfected cells,¹⁶ which was soon strengthened by direct evidence for homophilic binding and identification of the interacting domains.¹⁷ PECAM-1 was shown to contribute to the barrier function of endothelium¹⁸ as shown *in vitro*^{19–21} and by vascular permeability enhancing antibodies against PECAM-1 in mice.¹⁹ In addition, gene inactivation of PECAM-1 enhanced the induction of vascular permeability caused in mice by lipopolysaccharide-induced endotoxemia.^{22,23} Besides a role in junction integrity, PECAM-1 is involved in shear sensing²⁴ and the control of shear stress-induced atherosclerosis lesion development.^{25,26}

JAM-A, originally found as a platelet antigen,²⁷ was later identified as a founding member of the JAM family and a tight junction-associated protein of endothelial

Highlights

- Gene inactivation of ESAM (endothelial cell-selective adhesion molecule) enhances vascular permeability in the lung, whereas the heart, skin, and brain are unaffected.
- Deletion of JAM (junctional adhesion molecule)-A or PECAM (platelet endothelial cell adhesion molecule)-1 does not affect vascular barrier integrity in any of these 4 organs.
- Gene inactivation or blocking of VE-cadherin strongly enhances lung vascular permeability, yet rupturing of endothelial junction architecture leading to lethal blood coagulation is only seen if the gene for ESAM is simultaneously inactivated.

and epithelial cells.²⁸ The JAMs form mainly homophilic interactions,¹² and are involved in the transmigration of leukocytes, which has been summarized in several excellent reviews.^{15,29,30} The role of JAM-A for the stability and regulation of endothelial junctions is less well studied. Antibodies can slow down the reformation of destabilized junctions in transient calcium depletion assays with cultured ECs.³¹ This study also showed that anti-JAM-A antibodies caused corneal swelling in rabbits. For epithelium, the role of JAM-A for the stability of tight junctions was intensively studied, with dramatic defects in the intestinal barrier function in JAM-A^{-/-} mice³² and enhanced susceptibility to lipopolysaccharide-induced pulmonary edema and lung epithelial integrity in these mice.³³ A role of JAM-A in the regulation of epithelial tight junctions was also found in various human epithelial cell lines.^{34,35}

Like the JAMs, ESAM also contains 2 Ig domains and mediates homophilic interactions, although it is structurally less closely related to the JAMs than they are to each other. ESAM was identified as an endothelial antigen located at tight junctions, which is also expressed on platelets but not on epithelial cells.^{36,37} Although ESAM is generally not found on leukocytes, it is specifically expressed on hematopoietic stem cells.^{38–40} Targeted disruption of ESAM in mice (ESAM^{-/-}) did not show obvious developmental defects, yet pathological angiogenesis was affected, leading to reduced tumor growth.⁴¹ In zebrafish, distinct and redundant roles for ESAM were described in vascular morphogenesis.⁴² ESAM^{-/-} mice had no defects in the integrity of the endothelial barrier function in the skin or the peritoneum.⁴³ Unexpectedly, however, the increase of vascular permeability induced by inflammatory stimuli in skin and peritoneum was reduced in ESAM^{-/-} mice and also neutrophil extravasation was slowed down in the chemically inflamed peritoneum and the TNF (tumor necrosis factor)- α stimulated cremaster muscle.⁴³ A similar role in the context of vascular permeability induction was

found for JAM-C,⁴⁴ suggesting that ESAM and JAM-C are somehow involved in mechanisms that support the opening of junctions under inflammatory conditions.

Here, we show that deletion of ESAM, but not PECAM-1 or JAM-A, dramatically increased the junction destabilizing effect of VE-cadherin blocking antibodies. These effects were restricted to the lung and comprised complete physical destruction of endothelial junctions and immediate lethality, an effect, which was not seen by interfering with VE-cadherin alone. This was reproduced in double gene-deficient mice (ESAM^{-/-}/VE-cadherin^{IECKO}). We conclude that ESAM is essential for vascular integrity in an organ-specific way and acts additively with VE-cadherin.

MATERIALS AND METHODS

All data and supporting materials have been provided with the published article.

Cell Culture and Transfection

Primary ECs from lungs (mouse lung microvascular endothelial cells [MLMVECs]) of wild-type (WT) or ESAM gene inactivated mice were isolated and cultured as previously described.⁹ COS-7 (CV-1 in origin with SV40 genes) cells were cultured and transfected as described.⁴⁵ WT Chinese hamster ovary (CHO) cells, CHO cells expressing mouse JAM-A or ESAM were cultured as described.^{37,46}

Antibodies

The following antibodies were used: rabbit polyclonal antibodies (rab pAb) against mouse MAGI-1 (membrane-associated guanylate kinase, WW and PDZ domain containing 1; VE-18) and pAb against the extracellular domain of mouse ESAM (VD3) were raised and purified as described⁴⁷ and,³⁷ respectively. Rab pAb VE19 and rat mAb 1G8 both against ESAM,³⁷ rat mAb BV13 (Thermo Fisher Scientific), rab pAb VE42,⁴⁸ and goat pAb AF1002 (R&D Systems) against mouse VE-cadherin, rat mAb 1G5.1 and rat mAb 5D2.6 against PECAM-1,⁴³ rab pAb against JAM-A,⁴⁹ rat mAb against TER-119 (TER-119: BioLegend), rat mAb against GP (glycoprotein)Ib-β subunit of the mouse platelets (X649, Emfret Analytics), rab pAb against ZO-1 (zonula occludens-1; 40-2200, Thermo Fisher Scientific), rab pAb against Mesothelin (PA5-79698, Thermo Fisher Scientific), isotype control Ab (Rat IgG1, Thermo Fisher Scientific), mAb against α-tubulin (B-5-1-2, Sigma-Aldrich), and rab pAb against claudin-5 (34-1600 Thermo Fisher Scientific). Alexa Fluor 488–, Alexa Fluor 568–, Alexa Fluor 594–, and Alexa Fluor 647–coupled secondary antibodies were purchased from Invitrogen. All other secondary antibodies were purchased from Dianova.

Mouse Strains

The following mouse lines were used: 8- to 12-week-old C57BL/6Jrj WT mice (Janvier Labs), ESAM deficient mice,⁴³ PECAM-1 deficient mice,⁵⁰ and JAM-A deficient mice⁵¹ were used. ESAM^{-/-} Cdh5^{IECKO} or control ESAM^{-/-} Cdh5^{lox/lox} mice were generated by mating ESAM^{-/-} mice⁴³ with Cdh5^{lox/lox} mice⁹

either expressing PDGFbCre or not. ESAM-ΔSH3DBM and ESAM-mCherry knockin mice were generated by homologous recombination using a genetically modified targeting construct. The genomic sequence of ESAM was cloned from BAC-clone RP23-383h20 (Thermo Fisher Scientific). The knockin construct comprised an 8 kb genomic sequence from Exon 2 to Exon 7 (including the 1.3 kb mutagenesis region of Exon 7), an FRT (flippase recognition target)-site flanked hygromycin resistance gene, a sequence for polymerase chain reaction-genotyping and a homologous region of 1.7 kb containing the 3' untranslated region and Exon1 of the adjacent gene Vsig2. The ESAM-ΔSH3DBM mutation was generated using site-directed mutagenesis to delete the 3 potential SH3-binding motifs A₃₂₇APPRP₃₃₂, P₃₄₉RL₃₅₁, and M₃₈₃VP₃₈₅ in the cytoplasmic region of ESAM. To generate the ESAM-mCherry mutation, mCherry was fused to the C terminus of ESAM with the short linker AGLGG, thereby replacing the stop-codon of ESAM. The targeting construct was transfected into mouse embryonic stem cells of C57BL/6 mice using electroporation. Hygromycin-resistant colonies were screened by a polymerase chain reaction and confirmed by Southern blotting. Positive ES cell clones were injected into blastocysts of C57BL/6 mice to generate chimeras, which were mated with C57BL/6 mice. Genotyping was performed to verify the presence of the mutation. All animal studies were approved by the Landesamt für Natur, Umwelt und Verbraucherschutz Nordrhein-Westfalen. Animal studies were performed exclusively with male mice since they were abundant because of the use of female mice in another project.

Expression Vectors

mCherry was fused to the C terminus of ESAM with the linker GAGG and cloned into pcDNA3 vector for expression in COS-7. MAGI-1-FLAG expression vector was described.⁴⁷

Electron Microscopy

Samples were prepared as previously described⁹ with minor modifications. Mice were anesthetized with ketamine/xylazine, and tracheas were cannulated with an intravenous catheter tube. Lungs were perfused with PBS, followed by fixative buffer (2% paraformaldehyde and 2% glutaraldehyde in 0.1 mol/L cacodylate buffer, pH 7.2), then filled with fixative buffer via the trachea. Lungs were further fixed and processed as described.⁹

Immunofluorescence Staining

Staining of cultured cells and tissue vibratome sections were performed as described under Methods in the [online-only Data Supplement](#).

In Vivo Vascular Permeability Assay of Inner Organs

Mice were consecutively intravenously injected with rat IgG1 or BV13 antibodies and Evans blue and were euthanized 30 minutes later. For lung permeability assays, 6.5 hours after antibody application, Evans blue was administrated for 15 minutes. The body circulation was perfused, organs of interest were removed, and Evans blue was extracted with formamide for 5 days. The concentration of the dye was measured at 620 nm with a spectrophotometer (Shimadzu).

Cell Aggregation Assay

Assays were performed with ESAM and JAM-A transfected cells as described under Methods in the [online-only Data Supplement](#).

Electric Cell-Substrate Impedance Sensing

MLMVECs were seeded onto L-cysteine-reduced, fibronectin-coated 8W10E electrodes (Applied Biophysics). Electrical impedance was measured at 4000 Hz in real-time at 37°C and 10% CO₂ using the ECIS ZΘ system (Applied Biophysics). Confluent cells were treated with 50 µg/mL of control antibodies, BV13 or VD3 for 2 hours.

Immunoprecipitation and Immunoblotting

Coimmunoprecipitations and immunoblotting were performed as previously described⁵² except that the 4.5 µm Dynabeads (Thermo Fisher Scientific) were used instead of protein A/G-Sepharose.

Tamoxifen Treatment of Mice

Tamoxifen (100 mg) was mixed with 100 µL ethanol and 5 mL peanut oil and dissolved for 30 minutes in a water bath at 37°C using ultrasound. One hundred microliters of this solution was injected intraperitoneally daily for 5 consecutive days into ESAM^{-/-} Cdh5^{lox/lox} mice either expressing PdgfbCre or not, and mice were analyzed 2 or 3 days later.

Bone Marrow Transplantation and Platelet Isolation

Transplantations were performed as described.⁵³ Platelets were isolated as described,⁵⁴ then lysed with 2X SDS sample buffer and subjected to Western blotting.

FACS Staining of Heart and Lung ECs

Cell isolation and flow cytometry were performed as described under Methods in the [online-only Data Supplement](#).

Statistical Analysis

Data sets were tested for normality (Shapiro-Wilk) and equal variance (F test or Brown-Forsythe). Comparisons between 2 groups (1-factorial experiments) were performed using 2-tailed Student *t* test or Wilcoxon signed-rank test for normal and non-normally distributed data, respectively. Comparisons among 3 groups (1-factorial experiments) were performed using 1-way ANOVA followed by Tukey multiple comparison test or Welch ANOVA followed by Dunnett multiple comparison test for normally distributed data with either equal or unequal variances, respectively. Two-way ANOVA followed by Tukey multiple comparison test was used for 2-factorial experiments. GraphPad Prism8 software was used for this analysis. *P* values are indicated by asterisks: **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001. Results are shown as means ± SEM. Immunoblot signals were quantified using the software Image Studio (LI-COR) or Fiji (version 1.52f).

RESULTS

Effects of Blocking VE-Cadherin Function in ESAM^{-/-}, JAM-A^{-/-}, and PECAM-1^{-/-} Mice

We have previously shown that induced EC-specific deletion of the adhesion molecule VE-cadherin in adult mice (Cdh5^{IECKO} mice) increases vascular permeability selectively in the heart and lung.⁹ Despite this strong increase in vascular permeability in Cdh5^{IECKO} mice, ultrastructural analysis by electron microscopy revealed that endothelial junctions were not physically ruptured, suggesting that additional adhesion molecules maintain the junction structure in the absence of VE-cadherin.

This prompted us to investigate, which junctional molecules are relevant for the maintenance of junctional architecture despite the loss of VE-cadherin. As candidates, we analyzed the relevance of ESAM, JAM-A, and PECAM-1. To this end, we administered the anti-VE-cadherin antibody BV13⁸ or an isotype-matched control antibody intravenously into WT mice and ESAM^{-/-}, JAM-A^{-/-}, or PECAM-1^{-/-} mice, immediately followed by intravenous injection of Evans blue. After 30 minutes, mice were euthanized and extravasation of the albumin-adsorbed dye was measured in the lung, heart, skin, and brain. When comparing the different genotypes treated with control antibody, we found that the lack of PECAM-1 and JAM-A did not lead to any changes in basal permeability of any of the 4 organs (Figure 1A). In contrast, the absence of ESAM caused a 3-fold increase in vascular permeability in the lung and had no effect in the heart, skin, and brain (Figure 1A). When analyzing the effect of the anti-VE-cadherin antibody in WT mice, we found that vascular permeability was only increased in the lung and heart, but not in the skin and brain, in agreement with previous results.^{8,9} Accessibility of the intravenously injected antibody to VE-cadherin could be demonstrated as shown for brain capillaries (Figure 1B). The absence of JAM-A or PECAM-1 did not enhance the effect of BV13 in the lung or heart and did not enable the antibody to interfere with vascular integrity in the skin or brain. In contrast, gene inactivation of ESAM further enhanced the permeability-inducing effect of BV13 in the lung but had no destabilizing effect in other organs even in the presence of the anti-VE-cadherin antibody (Figure 1A).

We planned initially to administer the antibody BV13 and measure permeability after 2 hours, since effects on permeability are increasing between 1, 2, and 7 hours.⁸ However, we observed that between 35 and 45 minutes after injection of the antibody, ESAM^{-/-} mice showed strong dyspnea symptoms and died shortly afterwards. No such effects were seen in any of the other genotypes, clearly highlighting the synergistic function of ESAM and VE-cadherin, which together are essential for survival.

Although the vascular permeability enhancing effect in the lung of ESAM^{-/-} mice was slightly higher than the effect observed after 30 minutes BV13 treatment of WT mice,

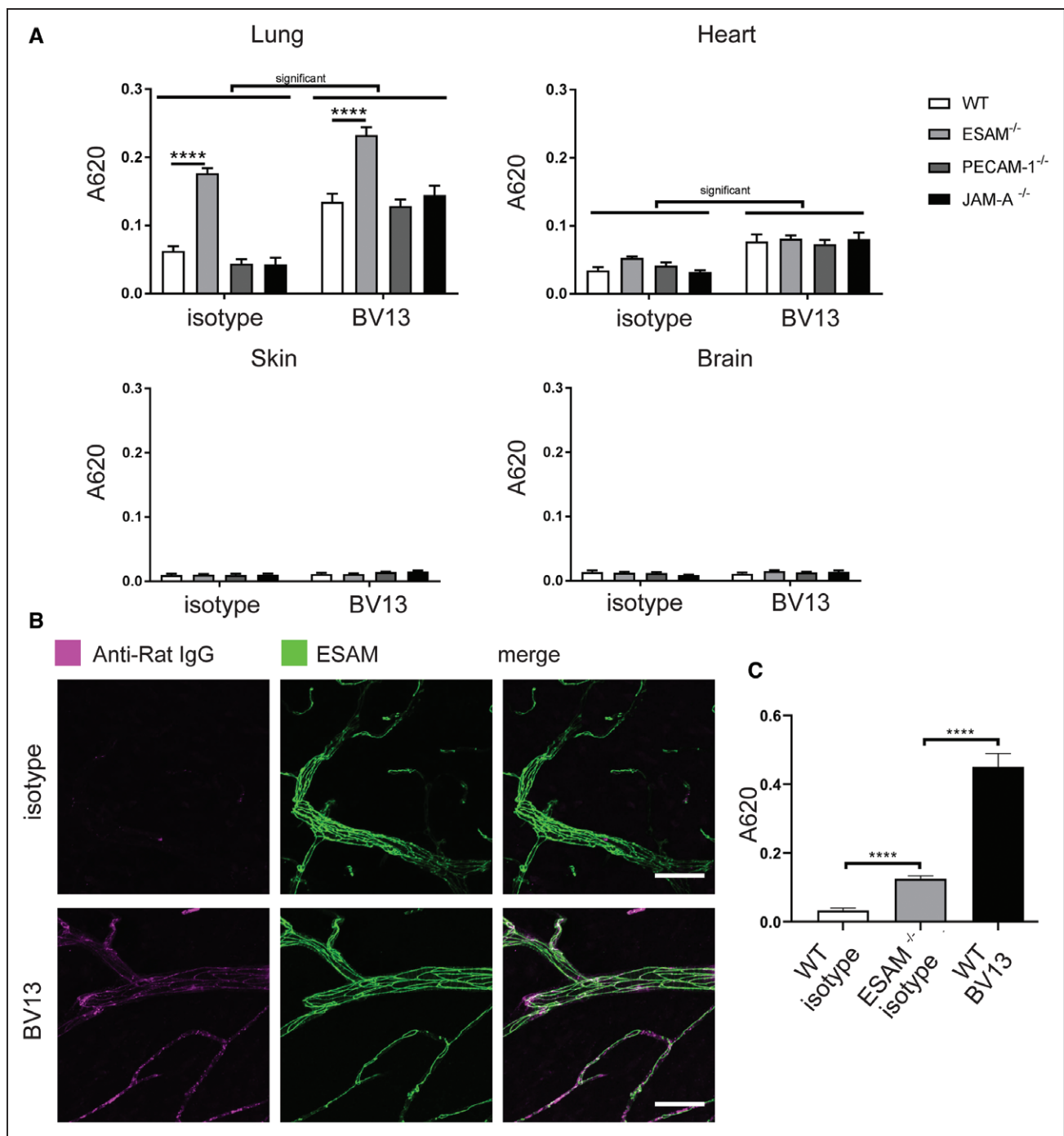


Figure 1. Vascular permeability measurements in different organs upon anti-vascular endothelial (VE)-cadherin treatment of ESAM^{-/-}, PECAM-1^{-/-}, or JAM-A^{-/-} mice.

A, Mice of various genotypes (as indicated) were intravenously injected with isotype control or anti-VE-cadherin antibody (BV13) and successively with Evans blue, and euthanized 30 minutes later. The vasculature was perfused and the dye was extracted and quantified from samples of various organs (as indicated). **B**, Brain vasculature was labeled by intravenous injection of isotype control antibody or rat antibody BV13, followed 20 min later by ex vivo incubation of 100 μ m brain vibratome sections with a secondary anti-rat IgG (magenta) and anti-ESAM (endothelial cell-selective adhesion molecule) antibodies (green). Bar: 50 μ m. **C**, Mice were intravenously injected with isotype control or BV13 antibodies. 6.5 h later, Evans blue was intravenously injected and 15 min later mice were euthanized, the body circulation was perfused, and the dye was extracted from lungs and quantified. Data are pooled from at least 2 independent experiments with 4 to 5 mice per group in each experiment (**A** and **C**) or representative of 2 independent experiments with 2 mice per condition (**B**). Statistical significance was analyzed using the 2-way ANOVA followed by Tukey test (**A**) or the Welch ANOVA followed by Dunnett test for normally distributed data with unequal variances (**C**). **** $P \leq 0.0001$. Results are shown as means $\pm$ SEM.

we do not conclude that ESAM is of similar importance for junction stability as VE-cadherin. First of all, ESAM^{-/-} mice are viable and seem to be able to compensate somehow for the higher vascular permeability in their lungs.^{41,43} Second, VE-cadherin function is far from being fully blocked at 30 minutes after BV13 injection, since the antibody takes some time to reach binding equilibrium.⁸ When we applied the antibodies for 6.5 hours, lung permeability in BV13 treated WT mice was almost 4 times higher than in control-IgG treated ESAM^{-/-} mice, showing that the loss of VE-cadherin function affects vascular integrity more severely than the loss of ESAM (Figure 1C). Nevertheless, our results clearly show that in the absence of ESAM, even mild, partial inhibition of VE-cadherin is sufficient to cause immediate lethality. This effect is not seen upon blocking VE-cadherin if ESAM is expressed at junctions.

Endothelial Junctional Integrity Requires Both VE-Cadherin and ESAM

Enhanced vascular permeability is one way of affecting junctional integrity, yet this is not necessarily accompanied by loss of physical intactness and rupture. We observed previously that Cdh5^{IECKO} mice showed strongly enhanced vascular permeability in the lung, but no signs of rupture of endothelial junctions.⁹ To investigate the physical state of the junctions in the different knockout mice challenged with BV13, we analyzed lung endothelial junctions by electron microscopy. We found that lung endothelial junctions of WT, JAM-A^{-/-}, and PECAM-1^{-/-} mice treated with isotype control antibodies were intact and even those of ESAM^{-/-} mice showed no signs of rupture or even partial damage (Figure 2A). Similar results were obtained when we blocked VE-cadherin with BV13 for 30 minutes in WT, PECAM-1^{-/-}, and JAM-A^{-/-} mice. Intriguingly, blocking VE-cadherin in ESAM^{-/-} mice led to the disruption of lung EC junctions, observed as gaps or open spaces between otherwise intact tails or trailers of ECs (Figure 2A). For quantification, randomly taken pictures of each genotype or treatment group were analyzed and intact junctions (electron-dense areas of EC contacts) plus gaps/open spaces between ECs (damaged junctions) were counted and set to 100%. This revealed that 58% of all junctions were damaged in BV13 treated ESAM^{-/-} mice, whereas no damage was found in the other groups (Figure 2C).

To exclude any potential side effects of the anti-VE-cadherin antibody, we generated mice double deficient for ESAM and VE-cadherin by mating ESAM^{-/-} mice with Cdh5^{lox/lox} mice expressing tamoxifen-inducible PdgfrbCre (Cdh5^{IECKO}). Mice expressing no Cre were used as negative controls. ESAM^{-/-} Cdh5^{IECKO} or ESAM^{-/-} Cdh5^{lox/lox} mice were stimulated with tamoxifen daily for 5 consecutive days and 2 to 3 days later, we collected the lungs and analyzed them by electron microscopy. In line with our blocking antibody data, we found that ablation of VE-cadherin together with ESAM led to disruption of endothelial junctions (Figure 2B). Quantification

(Figure 2D) showed that 56% of the junctions were damaged. In agreement with this, double gene-deficient mice did not survive >3 days after the tamoxifen injection period.

Collectively, our results show that ESAM is able to maintain the structure of endothelial junctions in the absence of VE-cadherin. Only when both proteins are absent from EC junctions, the junctional structure cannot be sustained, which leads to rupture and gaps between ECs.

Blocking or Deleting VE-Cadherin in ESAM^{-/-} Mice Leads to Lung Thrombosis

Since the combination of ESAM ablation and blocking or deleting VE-cadherin in ECs leads to lethality, rupture of junctions in the lung and exposure of the basement membrane, we investigated whether thrombus formation could be the cause for lethality. To this end, we injected control or BV13 antibodies into WT or ESAM^{-/-} mice. Thirty minutes later, mice were euthanized, the lung circulation was perfused and the tissue was fixed. Staining of lung vibratome sections for TER-119 and GPIb-V-IX revealed that erythrocytes and platelets accumulated in the lungs of BV13 treated ESAM deficient mice, but not in lungs of BV13 treated WT mice (Figure 3A). We validated the observation by electron microscopy, which showed clear signs of thrombosis at ruptured junctions which exposed freely accessible regions of the basement membrane (Figure 3B). Different stages of junction disruption and coagulation were found. Sites of loosened junctions without the typical electron-dense structure were probably representing early stages of junction rupture (first panel), whereas gaps between ECs (second panel) and finally formation of a thrombus containing activated platelets and erythrocytes (third panel) represented later stages of junction rupture and thrombus formation. Similar results were obtained from ESAM^{-/-} Cdh5^{IECKO} mice (Figure 3C).

We found previously that ESAM is not only expressed by ECs but also by platelets³⁷ and platelet ESAM was reported to limit thrombus growth and stability.⁵⁵ We, therefore, investigated whether the absence of ESAM on platelets contributed to the thrombus formation we observed in BV13 treated ESAM^{-/-} mice. To this end, we transplanted WT bone marrow into ESAM^{-/-} mice and into WT mice for controls. Platelets of the resulting mice were analyzed by immunoblotting for ESAM expression, documenting efficient repopulation of ESAM^{-/-} mice with WT platelets (Figure 1C in the [online-only Data Supplement](#)). The transplanted mice were treated with either BV13 or control antibody, followed by staining of vibratome sections of the lungs, as described above. We found that BV13 induced thrombus formation in ESAM^{-/-} mice transplanted with WT bone marrow similarly as we had found before for ESAM^{-/-} mice, whereas no thrombus formation was found in WT mice transplanted with WT bone marrow (Figure 1A in the [online-only Data Supplement](#)). Furthermore, we repeated Evans blue-based permeability assays (as in Figure 1A) and found similar

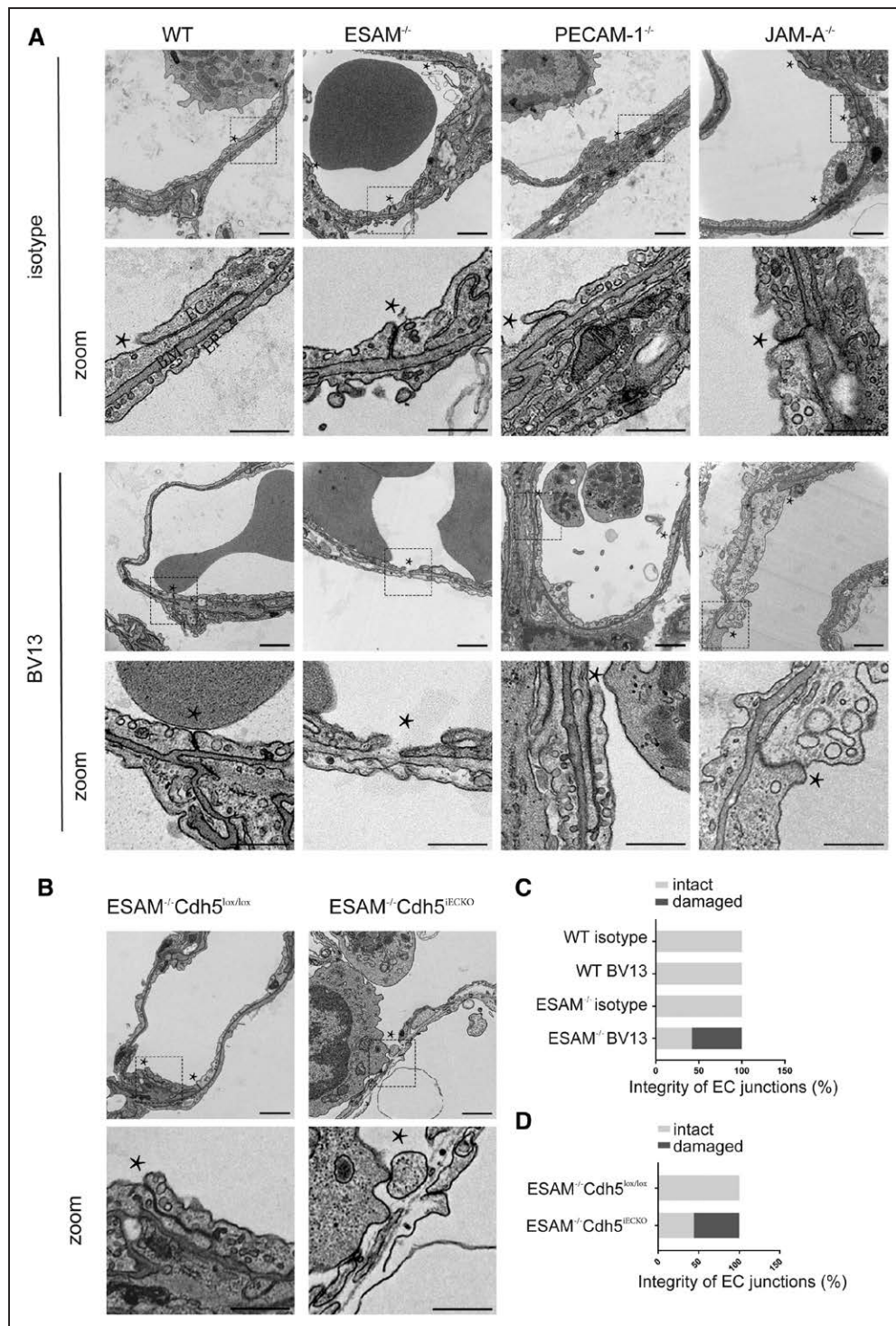


Figure 2. Blocking or deleting vascular endothelial (VE)-cadherin in ESAM^{-/-} mice leads to physical rupture of lung endothelial junctions.

A, Electron microscopy (EM) of lung endothelial cell junctions of isotype or BV13 antibody-treated mice. Thirty minutes after intravenous injection with isotype or BV13 antibodies, mice were euthanized, the lung circulation was perfused and fixed for further electron microscopy analysis. **B**, EM of tamoxifen-treated ESAM^{-/-} Cdh5^{lox/lox} and ESAM^{-/-} Cdh5^{ECKO} animals. Mice were intraperitoneally injected with tamoxifen daily for 5 d. Two to 3 days later, mice were euthanized, and lungs were prepared for electron microscopy. Asterisks mark junctions or openings. Bar: 1 μ m (original images) and 500 nm (zoom images). The ratios of damaged junctions to intact junctions were counted in electron microscopy pictures of lungs of **(C)** wild-type (WT) or ESAM^{-/-} mice treated with isotype or BV13 antibodies or **(D)** tamoxifen-treated ESAM^{-/-} Cdh5^{lox/lox} and ESAM^{-/-} Cdh5^{ECKO} mice. In total, 80 junctions in isotype treated WT, 85 junctions in BV13 treated WT, 59 junctions in isotype treated ESAM^{-/-}, 112 junctions in BV13 treated ESAM^{-/-}, 77 junctions in ESAM^{-/-} Cdh5^{lox/lox}, and 100 junctions in ESAM^{-/-} Cdh5^{ECKO} mice were counted. Data are representative of 2 independent experiments with at least 2 mice per condition. BM indicates basement membrane; EC, endothelial cell; and EP, epithelial cell.

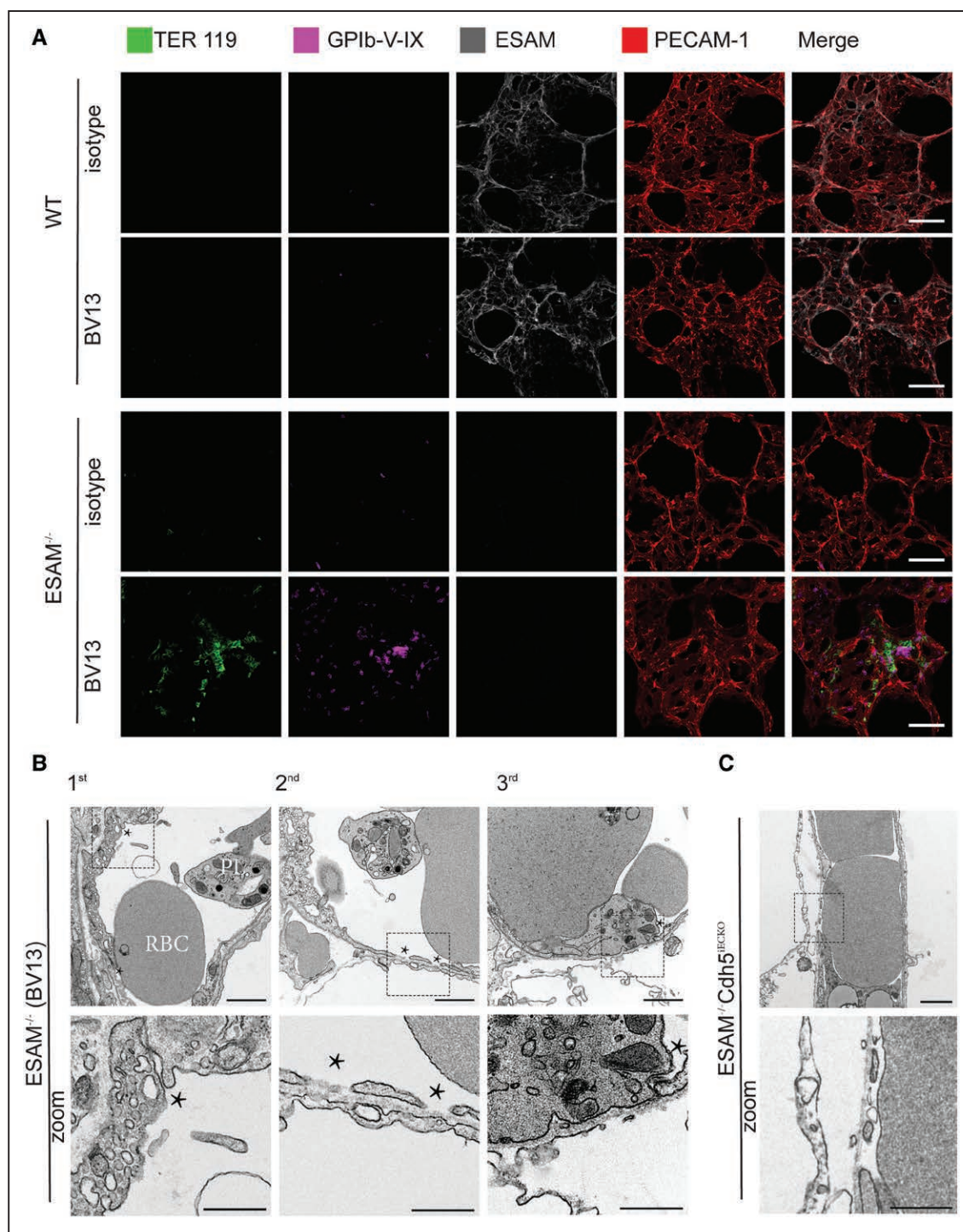


Figure 3. Blocking vascular endothelial (VE)-cadherin in ESAM^{-/-} mice leads to lung thrombosis.

A and **B**, Mice were intravenously injected with isotype or BV13 antibodies. Thirty minutes later, mice were euthanized, the lung circulation was perfused, fixed, and analyzed (**A**) by immunofluorescence microscopy of vibratome sections and (**B**) by electron microscopy (EM) of ultrathin sections. **B**, Different stages of junction rupture and thrombosis in ESAM^{-/-} mice treated with BV13 (from left to right): 1st, endothelial junctions preopening state; 2nd, endothelial junctions opening; and 3rd, thrombosis formation. **C**, Lung EM image of tamoxifen-treated ESAM^{-/-} Cdh5^{ECKO} mice. Asterisks mark junctions or openings. Bars: (**A**) 30 μ m; (**B** and **C**) 1 μ m (original images) and 500 nm (zoom images). Data are representative of 3 independent experiments with one mouse per condition (**A**) or 2 independent experiments with at least 2 mice per condition (**B** and **C**). ESAM indicates endothelial cell-selective adhesion molecule; PECAM, platelet endothelial cell adhesion molecule; PL indicates platelet; and RBC, red blood cell.

results with ESAM^{-/-} mice transplanted with WT bone marrow as before with ESAM^{-/-} mice (Figure 1B in the [online-only Data Supplement](#)). We conclude that blocking

or deleting VE-cadherin in ESAM^{-/-} mice induced lethal lung thrombus formation and enhanced vascular permeability in the lung independently of ESAM on platelets.

ESAM Expression Levels Are Higher in the Lung Than in the Heart

Whereas blocking or deleting VE-cadherin selectively enhances vascular permeability in the lung and heart, deletion of ESAM only has this effect in the lung. To understand the reason for this, we analyzed the

expression levels of ESAM in the lung and heart by immunoblotting. WT mice were euthanized, the vasculature was perfused and lungs and hearts were lysed and subjected to Western blotting. As shown in Figure 4A and quantified in Figure 4B, expression levels of ESAM (normalized to VE-cadherin) in the lung were almost 2-fold higher than in the heart.

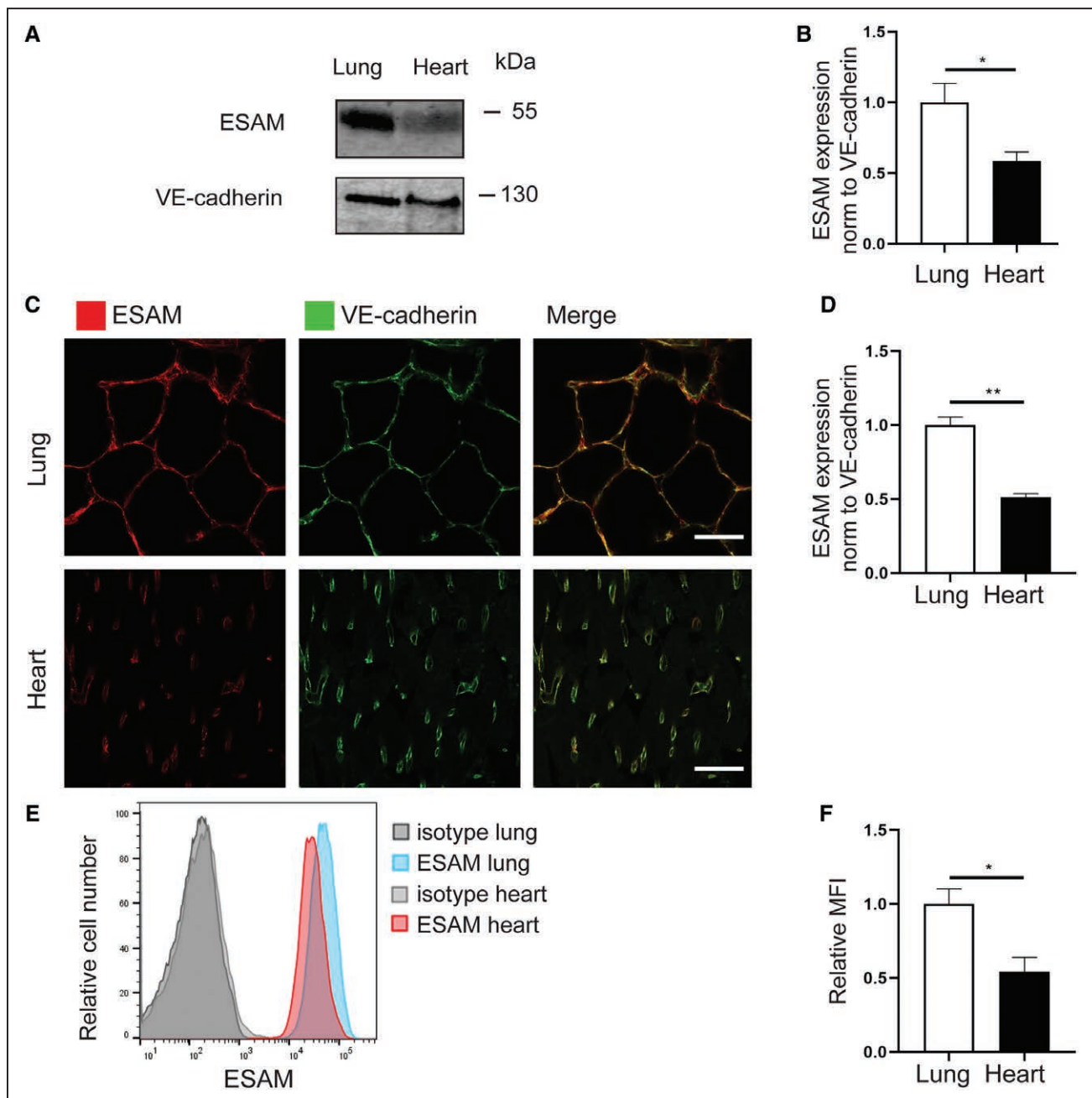


Figure 4. ESAM (endothelial cell-selective adhesion molecule) expression is higher in the lung than in the heart.

Wild-type (WT) mice were euthanized, and the body circulation was perfused, then hearts and lungs were collected. **A**, The organs were lysed and immunoblotted for ESAM and vascular endothelial (VE)-cadherin. **B**, Expression levels of ESAM (normalized to VE-cadherin) in hearts and lungs were quantified from immunoblot signals in **(A)** ($n=4$). **C**, Vibratome sections of mouse lung and heart were analyzed by immunofluorescence microscopy. Bar: 30 μm . **D**, Expression levels of ESAM in hearts and lungs were quantified from **(C)** (normalized to VE-cadherin) ($n=3$). **E**, Endothelial cells isolated from lungs and hearts were gated for VE-cadherin⁺ and CD31^{high} expression by fluorescence-activated cell sorter (FACS) analysis and analyzed for ESAM expression levels. **F**, Shows quantification of mean fluorescence intensity (MFI) signals from **(E)**; ($n=3$). Statistical significance was analyzed using the Student *t* test for normally distributed data with equal variances (**B**, **D**, and **F**). * $P \leq 0.05$; ** $P \leq 0.01$. Results are shown as means $\pm$ SEM.

To verify these results by another independent technique, we stained vibratome sections of lungs and hearts from perfused mice for ESAM and VE-cadherin (Figure 4C). In agreement with the previous immunoblot data, ESAM expression levels (normalized to VE-cadherin) in lungs were ≈ 2 times higher than in hearts (Figure 4D). In the course of these immunofluorescence experiments, we found that ESAM was not only restricted to ECs but was also expressed in mesothelial cells which surround the outer surface of inner organs such as lungs and hearts (Figure II in the [online-only Data Supplement](#)). Therefore, all images used to quantify ESAM expression in hearts and lungs were captured in the parenchyma of those organs to avoid the mesothelium.

As a third method, we compared the expression level of ESAM in the vasculature of lungs and hearts by fluorescence-activated cell sorter (FACS) analysis of primary isolated ECs. We determined ESAM expression levels by antibody staining of ECs which were gated as VE-cadherin⁺ and CD31^{high} cells. Quantification of mean fluorescence intensity revealed again a 2-fold higher expression level of ESAM on lung EC compared with heart ECs (Figure 4E and 4F). The higher expression level of ESAM in lung endothelium may partially explain why ESAM contributes more to the stability of endothelial junctions in the lung than in the heart.

ESAM Deletion Does Not Alter mRNA and Protein Expression of Other Junctional Molecules

Since endothelial adhesion molecules such as VE-cadherin support the expression of claudins,⁵⁶ we tested whether ESAM gene inactivation might affect the expression of other junctional molecules, which could be the reason for enhanced vascular permeability in the lung. To test this, we compared lung primary ECs (MLMVECs) isolated from WT and ESAM^{-/-} mice for the expression of VE-cadherin, JAM-A, claudin-5, and ZO-1 by immunoblots and immunofluorescence staining. As shown in Figure 5A and 5B, immunoblots showed no difference between these cells in the expression levels of these proteins. Likewise, their expression levels at cell junctions were unaffected by the disruption of the ESAM gene (Figure 5C).

To verify these results with primary isolated ECs (avoiding culturing of these cells), we next investigated the overall expression levels of additional junctional molecules at both the mRNA and protein level. We freshly isolated ECs from lungs of WT and ESAM^{-/-} mice by FACS sorting. Immediately after isolation, RNA was isolated from these cells and mRNA levels of 11 junctional molecules (JAM-A, -B, and -C; claudin-1, -3, -5, and -12; occludin; ZO-1; PECAM-1; and Cdh5) were analyzed by quantitative real-time polymerase chain reaction. In line with our results above, mRNA levels of each

of these molecules were similar for both genotypes (Figure 5D). Furthermore, staining of lung vibratome sections showed that in situ distribution and expression levels of VE-cadherin and claudin-5 were indistinguishable between WT and ESAM^{-/-} mice (Figure 5E).

Thus, we found no evidence for a reduction of junctional proteins in lung endothelium devoid of ESAM. This suggests that the loss of junction integrity we observed in the lung of ESAM^{-/-} mice is directly due to the absence of this adhesion molecule.

Deleting SH3 or Inactivating PDZ-Binding Motifs of ESAM Does Not Affect Vascular Lung Permeability

ESAM is a homophilic cell adhesion molecule.^{36,37} In addition, it contains putative binding motifs for SH3DBM (SH3-domains) and PDZDBM (PDZ-domains) in its cytoplasmic part.^{36,47} To test whether any of these sites might be relevant for endothelial junction integrity, we generated knockin mice, carrying mutations affecting these putative signaling sites. For the mutation of the 3 presumptive SH3DBM (Figure 6A), coding regions for the respective amino acids were deleted in a construct for homologous recombination (Figure 6C) which was used to generate knockin mice (see Materials and Methods section). The PDZDBM of ESAM is located at the C-terminal end of the protein and binds to the PDZD protein MAGI-1.⁴⁷ This binding motif becomes masked if the fluorescent protein mCherry is fused to its C terminus (Figure 6A), as we tested in coprecipitation experiments with COS-7 cells transfected with MAGI-1 and either full-length ESAM or ESAM-mCherry (Figure 6B). Since mCherry blocked the function of the PDZDBM, we decided to generate ESAM-mCherry knockin mice by homologous recombination (see Materials and Methods section) and analyze them for the relevance of the PDZDBM for vascular integrity.

The 2 knockin mouse lines ESAM- Δ SH3DBM and ESAM-mCherry were healthy and viable. Performing immunofluorescence staining for ESAM of vibratome sections of the lung, heart, skin, and brain, of WT mice and the 2 knockin mouse lines, we found that the expression levels and junction localization of the 2 modified versions of ESAM were indistinguishable from WT ESAM (Figure III in the [online-only Data Supplement](#)). To test whether the SH3DBM or PDZDBM contributes to endothelial junction integrity in lung, heart, skin, and brain, we performed Evans blue-based vascular permeability assays in these mice as described above. As shown in Figure 6D, none of the ESAM- Δ SH3DBM and ESAM-mCherry mutant mice showed a different level of vascular permeability than WT mice. We conclude that the increased vascular permeability we observed in ESAM^{-/-} mice is not due to impaired signaling functions of ESAM via its PDZ domain-binding site or its putative SH3DBM motifs.

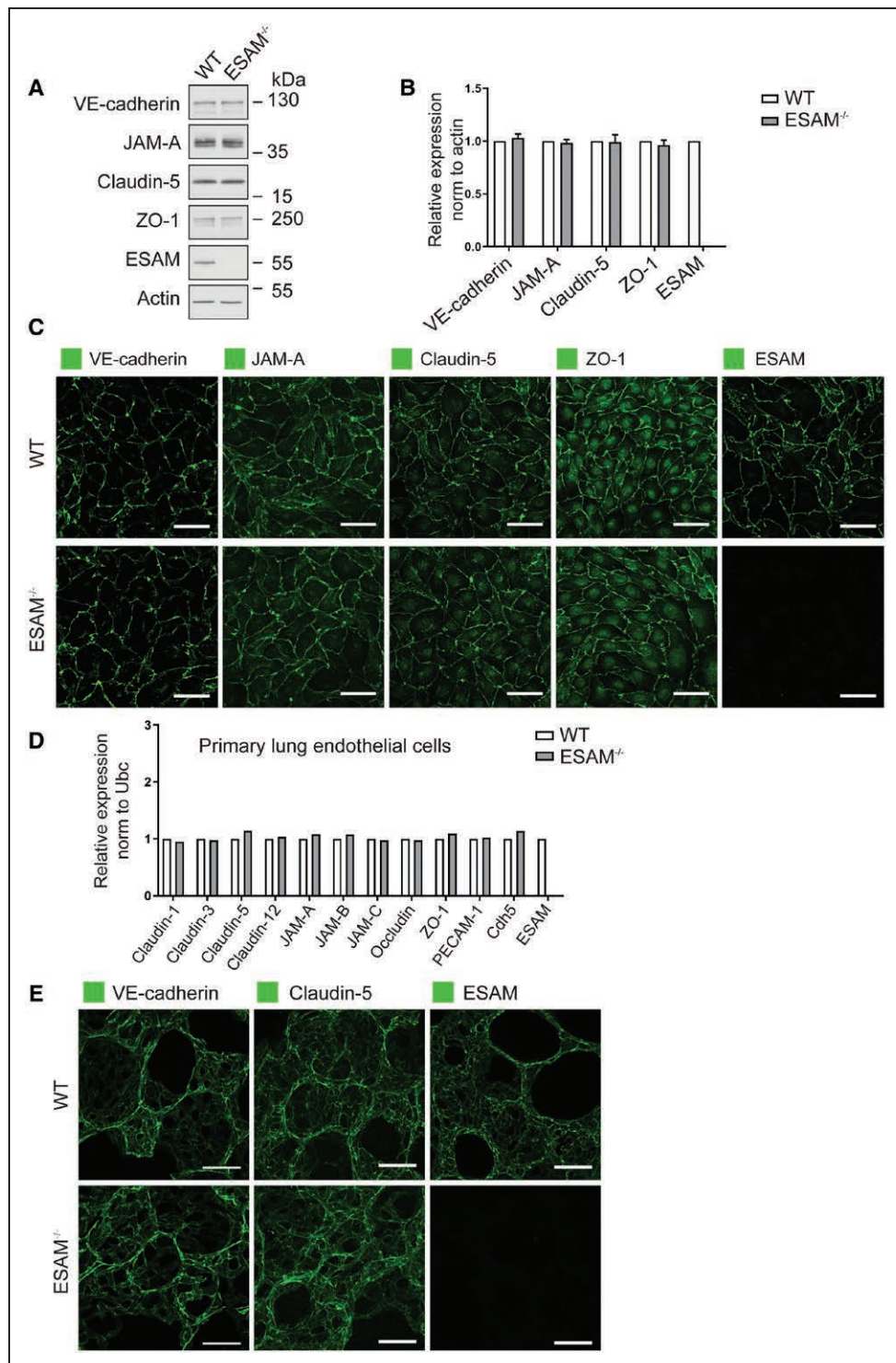


Figure 5. No differences in mRNA or protein expression of junctional molecules in lungs of ESAM^{-/-} mice compared with wild-type (WT) mice.

Mouse lung microvascular endothelial cells (MLMVECs) were isolated from WT and ESAM^{-/-} mice via bead-sorting with anti-CD31 antibodies and analyzed by **(A)** immunoblots for indicated antigens, **(B)** quantification of blot signals (n=5) or **(C)** staining in culture for indicated antigens (n=3). **D**, MLMVECs of control and ESAM^{-/-} mice double positive for vascular endothelial (VE)-cadherin and CD31^{high} were isolated by fluorescence-activated cell sorter (FACS) sorting and total RNA was extracted immediately after sorting and was analyzed for expression of junctional molecules by quantitative real-time polymerase chain reaction (qRT-PCR) using specific TaqMan probes. Samples were measured in triplicate, and diagrams are representative of RNA from 4 WT and 4 ESAM^{-/-} animals. Relative expression of target genes was normalized to the control transcript Ubc. **E**, Mice were anesthetized, euthanized, the lung circulation was perfused, fixed and lung vibratome sections were analyzed by immunofluorescence microscopy (n=3). Bars: **(C and E)** 30 μm. Statistical significance was analyzed using the Wilcoxon signed-rank test for non-normally distributed data **(B and D)**. ESAM indicates endothelial cell-selective adhesion molecule; and JAM, junctional adhesion molecule.

ESAM Contributes to Endothelial Junction Integrity by Mediating Adhesion

Next, we tested whether ESAM mediated adhesion could contribute to endothelial junction integrity. ESAM and JAM-A belong to the same subfamily of the large class of Ig-SF membrane proteins, which share a similar domain organization with 2 Ig-like domains, a single transmembrane domain and a cytoplasmic tail that ends in a canonical PDZ domain-binding sequence.¹² Both proteins were described as homophilic adhesion molecules, with ESAM mediating cell aggregation³⁶ and JAM-A being involved in the reformation of endothelial junctions after dissociating them by Ca^{2+} chelating reagents.^{31,57} Since we found that only ESAM^{-/-} mice but not JAM-A^{-/-} mice exhibited an increase in lung vascular permeability, we decided to directly compare the adhesion properties of both adhesion molecules in cell aggregation assays. To this end, we stably transfected CHO cells with ESAM or JAM-A and isolated cell clones expressing either ESAM or JAM-A at similar levels (Figure 6E). After 1 hour of incubation, ESAM-CHO cells were significantly aggregated when compared with untransfected CHO cells (Figure 6F). In contrast, JAM-A-CHO cells aggregated only slightly, but insignificantly better than CHO cells (Figure 6F). Thus, ESAM mediates cell contact formation and supports cell contacts more efficiently than JAM-A.

If ESAM directly contributes to EC contact stability by mediating homophilic cell adhesion, anti-ESAM antibodies should be able to interfere with endothelial junction integrity. Indeed, we found that electrical resistance of MLMVEC monolayers was reduced by 50 $\mu\text{g/mL}$ polyclonal antibodies (incubated for 2 hours) against the extracellular part of ESAM, while negative control antibodies from the corresponding preserum and antibodies against endomucin had no such effect (Figure 6G). In line with our *in vivo* results (Figure 1C), the mAb BV13 against VE-cadherin reduced electrical resistance across MLMVEC monolayers more strongly than the anti-ESAM antibodies (Figure 6H).

Collectively, our results suggest that ESAM could support endothelial junction integrity directly by its adhesive function. Although its contribution to junction stability is less dominant than the contribution of VE-cadherin (Figures 1C and 6H), the absence of ESAM causes immediate lethality even if VE-cadherin is only partially blocked.

DISCUSSION

Here, we have analyzed and compared the relevance of ESAM, JAM-A, and PECAM-1 for vascular integrity in various organs. In addition, we wanted to know whether blocking of the adhesive function of VE-cadherin would have additive effects, if any of these 3 adhesion molecules was deleted *in vivo*. We found that gene inactivation of JAM-A or PECAM-1 did not impair vascular integrity in any of the analyzed organs independent of whether VE-cadherin was fully functional or blocked with antibodies.

In contrast, gene inactivation of ESAM enhanced vascular permeability specifically in the lung, and not in the heart, brain, and skin, even when VE-cadherin was fully functional. Combined interference with ESAM and VE-cadherin had a dramatic additive effect on vascular integrity in the lung, whereas the VE-cadherin blocking effect in the heart was not enhanced and the brain and skin were unaffected. Thus, only in the lung, the lack of ESAM strongly enhanced the damaging effect of anti-VE-cadherin antibodies on vascular barrier function. This defect was so dramatic that endothelial junctions in lung capillaries were physically disrupted within 30 minutes, which led to massive blood clotting and immediate lethality, an effect that was not seen upon blocking VE-cadherin in WT mice.

It is well documented that interference with VE-cadherin disrupts endothelial junctions of cultured ECs, no matter from which organs they originate.^{2,58} However, fully established endothelial junctions in adult organs seem to be less dependent on VE-cadherin once they are established.⁹ Gene inactivation of VE-cadherin only increased vascular permeability in the lung and heart, but not in the skin or brain.⁹ Even more important, in none of these organs, gene inactivation of VE-cadherin caused physical disruption of endothelial junctions.⁹ This reveals, first that increase in permeability of macromolecular substances does not require physical rupture of junctions. Second, the physical stability of endothelial junctions can be maintained even in the absence of VE-cadherin.

Here, we have identified ESAM as the adhesion molecule that is responsible for the remarkable stability of lung endothelial junctions in the absence of VE-cadherin. This is based on 2 lines of evidence. First, blocking of VE-cadherin with antibodies physically disrupts the structure of endothelial junctions within 30 minutes and triggers rapid lethality in ESAM^{-/-} mice, but not in WT mice. This led to gap formation between ECs which triggered thrombus formation due to basement membrane accessibility to platelets. Second, endothelial junction rupture and lethality was detected in double gene inactivated mice again at an early stage at 7 to 8 days after the first day of tamoxifen administration, in contrast to a survival time of at least 3 weeks (mice were then euthanized) after induction of VE-cadherin gene disruption in ESAM-expressing mice. Thus, ESAM in the lung is essential and sufficient to maintain endothelial junctions in the absence of VE-cadherin and to prevent the formation of gaps between ECs.

The selectivity with which we observed rupture of endothelial junctions only in the lung was striking. However, it is difficult to exclude a potential additive role of VE-cadherin and ESAM for endothelial junction stability in other organs, because the dramatic defects of blocking VE-cadherin in ESAM^{-/-} mice in the lung vasculature cause thrombus formation and subsequent lethality within 30 minutes, thereby preventing the analysis of

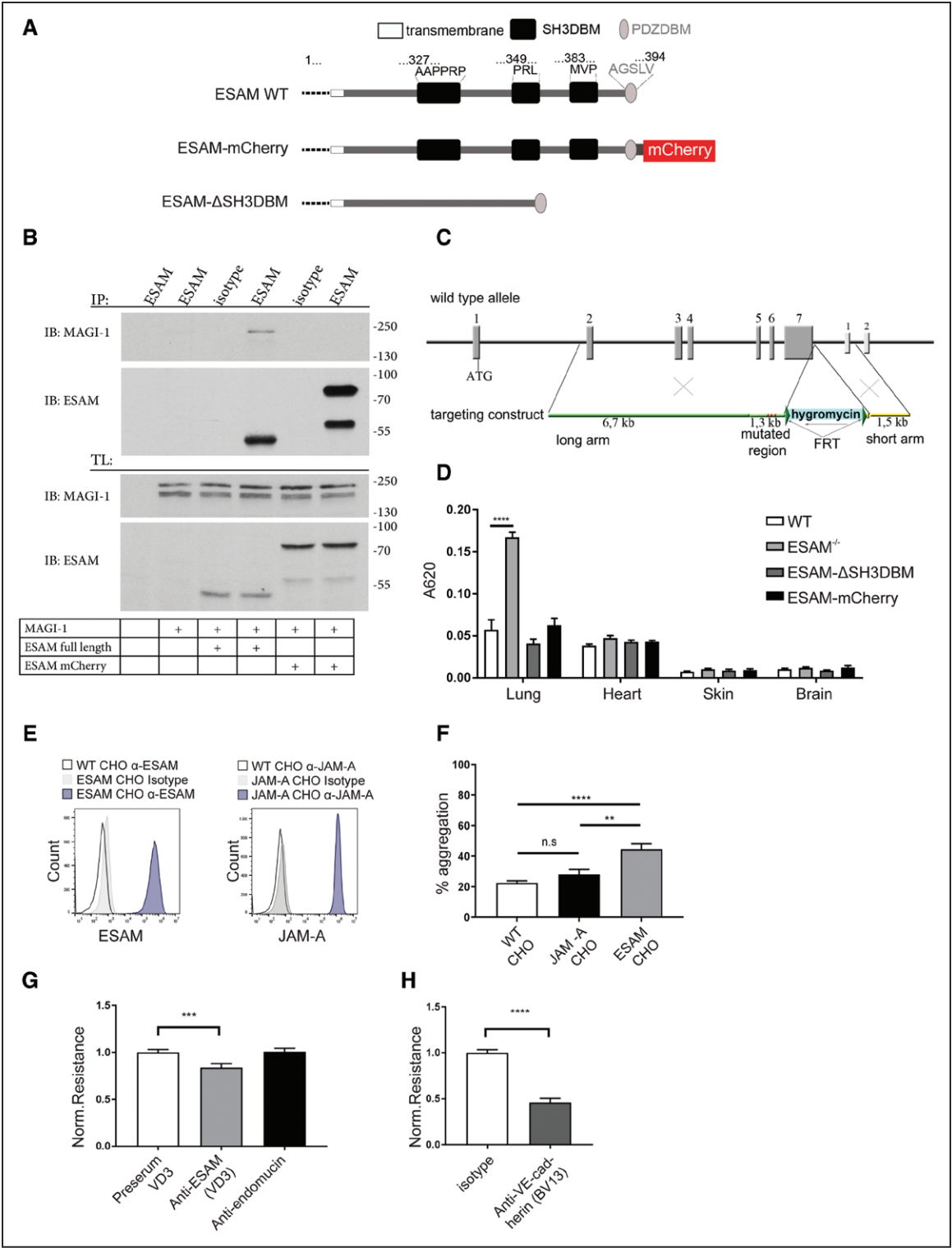


Figure 6. Adhesion properties of ESAM (endothelial cell-selective adhesion molecule), but neither its SH3DBM (SH3-domains binding motifs) nor PDZDBM (PDZ-domain-binding motifs) contribute to endothelial junction integrity. **A**, Schematic illustration of the location of the SH3DBMs and the PDZDBM within the cytoplasmic part of ESAM and of the corresponding genetic modifications of ESAM in the 2 knockin mouse lines (numbers refer to amino-acid positions). **B**, Cos-7 cells were cotransfected with different combinations of MAGI-1-FLAG, ESAM full-length or ESAM-mCherry (as indicated below) and either anti-ESAM immunoprecipitates (top) or total cell lysates (bottom) were immunoblotted for MAGI-1 or ESAM. Molecular mass markers are indicated on the right. **C**, Targeting strategy for the generation of ESAM knockin mouse mutants. **D**, Wild-type (WT) or mutated ESAM knockin mice (as indicated) were intravenously injected with Evans blue and euthanized 30 min later. The body circulation was perfused, the dye was extracted from samples of various organs (as indicated) and quantified. **E**, Expression of ESAM and JAM (junctional adhesion molecule)-A on transfected Chinese hamster ovary (CHO) cells determined by fluorescence-activated cell sorter (FACS). (Continued)

Figure 6 Continued. **F**, Cell aggregation assays with untransfected, JAM-A- or ESAM-transfected CHO cells, for 60 min in suspension, followed by fixation and counting of cell aggregates. **G** and **H**, Electrical resistance assays (monitored by ECIS) with mouse lung microvascular endothelial cells (MLMVEC) upon treatment with preserum IgG, anti-ESAM (VD3) or anti-endothelial (G); or isotype or anti-vascular endothelial (VE)-cadherin (BV13) antibodies (**H**) for 2 h. Data are representative of 3 independent experiments (**B** and **E**), pooled from 2 independent experiments with 4 to 5 mice per group in each experiment (**D**), or pooled from 3 independent experiments with 2 to 3 samples per group (**F**, **G**, and **H**). Statistical significance was analyzed using the 2-way ANOVA followed by Tukey test (**D**), the Welch ANOVA followed by Dunnett test for normally distributed data with unequal variances (**F**), the 1-way ANOVA followed by Tukey test for normally distributed data with equal variances (**G**) or the Student *t* test for normally distributed data with equal variances (**H**). ** $P \leq 0.01$; *** $P \leq 0.001$; and **** $P \leq 0.0001$. Results are shown as means $\pm$ SEM.

milder junctional defects in other organs, which might require some hours to detect. Nevertheless, an essential role of both adhesion molecules for junction integrity in the lung vasculature is obvious. A possible reason for the lung selectivity of vascular junction damage could be related to the permanent mechanical challenge of the breathing lung. In addition, the fact that ESAM is expressed at 2-fold higher levels at junctions of the lung vasculature when compared with vessels in the heart could be another reason why interference with ESAM and VE-cadherin had more dramatic effects in the lung. It is also possible that in other organs than the lung, it may be a multitude of adhesion molecules which provide junction stability in the absence of VE-cadherin.

ESAM has also been analyzed during segmental artery formation in zebrafish embryos. Filopodial contact formation during anastomosis was affected in VE-cadherin mutants, which were aggravated in VE-cadherin/ESAM double mutant embryos.⁴² In addition, ZO-1 staining of endothelial junctions was occasionally interrupted in sprouting segmental arteries in the absence of ESAM, which was found more frequently when both adhesion molecules were missing.⁴² Ultrastructural analysis of junctions was not performed.

JAM-A and PECAM-1, in contrast to ESAM, were each dispensable for physical maintenance of endothelial junctions in the lung, even if VE-cadherin function was blocked. As all 3 proteins (JAM-A, PECAM-1, and ESAM) are homophilic cell adhesion molecules, the question arises why ESAM is more important than the others for lung vascular integrity. This is especially interesting in the light of the important role of JAM-A for epithelial junction stability in vivo, as was documented for the intestine.³² Our side by side comparison of the ability of ESAM and JAM-A to support cell-cell aggregation of transfected cells revealed that ESAM is the more efficient cell adhesion molecule in terms of forming cell-cell contacts that are sufficiently robust to resist gentle pulling forces under rotation. In addition, antibodies against ESAM could interfere with the electrical resistance of cultured EC monolayers, further arguing that ESAM mediated adhesion supports endothelial junction integrity. Finally, the fact that lung endothelium expresses ESAM at 2-fold higher levels than, for example, heart capillaries again suggests that ESAM is required as a direct supporter of adhesion strength at endothelial junctions in the lung.

Although indirect contributions of ESAM to the physical strength of lung endothelial junctions cannot be excluded,

we could rule out any relevance of the PDZDBM at the C terminus and the 3 putative SH3DBM sites in the C-terminal part of ESAM, by analyzing appropriately mutated knockin mice. Analyzing mice with a deletion of the complete cytoplasmic part of ESAM was not possible since such a mutated form of ESAM is not expressed at the EC surface (unpublished results). Another indirect way to affect junction stability through the absence of an adhesion molecule would be by reducing the expression of other junctional molecules. Indeed, VE-cadherin has been shown to support the transcription of claudin-5.⁵⁶ For ESAM, we could exclude that its absence causes a decrease in the expression of most of the known endothelial junctional adhesion molecules, again strengthening the assumption that ESAM supports junction stability directly.

Despite this direct adhesive support of endothelial junction stability, we found previously that ESAM can indirectly contribute to mechanisms that destabilize endothelial junctions. We found that inflammation-induced vascular permeability in skin and peritoneal cavity was reduced in ESAM^{-/-} mice.⁴³ This argued for a role of ESAM in inflammation-induced signaling events that trigger the destabilization of endothelial junctions. Thus, signaling triggered by ESAM rather supports endothelial junction destabilization, in contrast to the adhesive function of ESAM, which is needed for steady-state junction stability.

In conclusion, we have shown that ESAM, but not JAM-A or PECAM-1, is essential for the barrier function of endothelial junctions in the lung. Furthermore, ESAM is able to prevent physical rupture of junctions in lung capillaries even if VE-cadherin is blocked or absent. Finally, only simultaneous interference with VE-cadherin and ESAM leads to rupture of the endothelial junctions of lung capillaries and rapid lethality, documenting the additive role of these adhesion molecules for endothelial barrier integrity.

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Disclosures

None.

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